R44: source orientation obstructs ordinary transfers

Observations, exact certificates and conditional geometric deductions. No independent evidence verdict, IC efficiency claim or larger-n transfer.

filter83-01/filter284-01: five generator blocks per form; no sampled success rate or IC rank.

Full source algebras

The scalar C5 color distinction from R30 did not determine the sign involution on the full source. R44 constructs actual source algebras for both degree-12 middles and the degree-436 middle, retaining the full degree-5 eleven-kernel and both orientation forms. All five original equations and eight unit inverses pass at all 10diamond iterates; exact Tate order 11 and corrupted-nu controls pass.

Base degree	Middle choices	Actual source tuple degree	F32 component degree
12	2	120	24
436	1	4360	872

Dimensions 5/10/20 are relative to the original coefficient field. The degree-4360 source degree uses actual b,c field generation of degree 2180 and AS irreducibility; it is not inferred solely from an ambient algebra dimension. This remains the n=5 carrier: a large ordinary residue degree is not a larger Koblitz n.

Exact sign obstruction

For d=12 or 436, actual source Frobenius satisfies F raised to 5d equals the sign sigma=T raised to 5. Thus sigma is a power of the F32 Frobenius and fixes each F32 closed source divisor. The mixed expression E=(1−sigma)(5−N5) therefore kills each such component, including individually colored ones. The proper/divisor interpretation uses the retained moduli/normal-curve bridge.

Trace screen, verification and remaining scope

The actual normalized orientation constant has an exact decomposition C=c2+delta, delta=b/A²+r=b/A²+b/A³, A=1+c. Here c2 is the original 7-kernel quadratic coefficient. Separate native traces give 1=1+0: source orientation 1, Tate-model twist 1, seven-kernel 0 in all three fixed middle cases.

This gives a concrete eligibility screen before expensive divisor work. All 20 retained 83/284 blocks reproduce their split flags: only the83 diamond twist has trace 0. Trace 0 does not prove a nonzero class or independent column. That 83 twist orbit already has R33/R37/R39 class data; it is not a new relation.

Controls and formal scope

Quotient-first evaluation agrees with all 52 direct values on four accepted source algebras. Matrix inverses agree with all 32 old inverses and recheck every original matrix row/product. Certified periods agree with the exhaustive old degree-12scan. The AS Frobenius recurrence matches all 16 direct coordinate values and all four sign comparisons. Full degree-436 source checks retain all branches, all ten diamond comparisons and actual quotient-map/sign certificates.

Lean checks seven named mathematical theorems plus four generated equations: sign/mixed cancellation, cleared orientation identity, quadratic Frobenius recurrence and odd constant descent. Zero failing/trusted. Native arrays, field generation and proper geometry are not formalized by those generic lemmas. Independent semantic review and scientific replay remain pending.

Failures and boundaries

Three full degree-436 implementations timed out at 1200 seconds; none is a mathematical refutation. A reserved Singular name and a nested-list control error were strictly rejected despite late markers. An unavailable Lean import was fixed in a fresh attempt. The third saved full geometry and first Frobenius; a separate receipt completes only the missing twisted Frobenius check. All attempts remain partial/rejected as appropriate, no completed block was regenerated and no cap increased. Native 4 GiB/1200 s; pinned Lean 4.34.1/mathlib d13f23b, one job, 4096 MiB heap, 1200 seconds compile/audit.

The base 367/460 labels are not source divisor degrees. This sign certificate eliminates these ordinary mixed contributions under the retained source binding; it does not complete every principal divisor or reverse image. Next screen remaining 500/580 families, then evaluate actual surviving classes. Proper target binding, full quotient saturation/polarization, theta and node187 remain separate. No IC relation rank, unknown-scalar recovery, performance comparison or evaluated larger-n transfer is asserted.

Full exact data/failure index: results.json; source/format links: LITERATURE.md and FORMAT.md; remaining work: NEXT.md. Artifact byte custody is checked separately from mathematical replay.